

choptuik_ac_bc · theorems12 · github.com/wild8highlander/choptuik_ac_bc

The Machine-Verified Hilbert Derivation: from the action to the double-null system

Hilbert identity exact · Roberts residual 4×10^{-41} · the whole system in one SymPy pass (1.9 s)

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Verdict:	einstein_direct/results/ derivation_results.json
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Abstract

This supplementary monograph records the machine-verified Hilbert derivation theorem: the whole double-null Einstein–scalar system of the repository is derived symbolically from the Hilbert action in one SymPy pass (1.9 s). The metric $ds^2 = -\alpha(u,v)^2 du dv + r(u,v)^2 d\Omega^2$ and the action $S = (1/2\kappa)\int\sqrt{-g} R + S_m$ with $S_m = -1/2\int\sqrt{-g}(\nabla\Phi)^2$ yield the Hilbert stress-energy $T_{uv} = 0$, $T_{uu} = \Phi_u^2$, $T_{vv} = \Phi_v^2$, the d'Alembertian $\Box\Phi = 4(-r\Phi_{uv} - \Phi_u r_v - \Phi_v r_u)/(\alpha^2 r)$, and the Hilbert identity $\Box\Phi - v = (\kappa/2)T - v$ holds exactly in both components.

The solved system matches the solver's equations term by term: $C1_{r_{uu}} = -\kappa r\Phi_u^2/2 + 2\omega_u r_u$, $C2_{r_{vv}}$ symmetric, $UV_{r_{uv}} = (-e^{2\omega}/4 - r_u r_v)/r$, $TH_{\omega_{uv}} = -\kappa\Phi_u\Phi_v/2 - r_{uv}/r$. The checks close the loop: $R_{\text{phiphi_spherical}} = \text{true}$, $\text{roberts_solution_ok} = \text{true}$ with the residual 4.06×10^{-41} , $\text{burko_E1_ok} = \text{true}$. Tables of the derivation, corollaries, honest caveats and the reproduction protocol are included.

Keywords: Hilbert action, symbolic derivation, Einstein–scalar system, double-null coordinates, Hilbert identity, Roberts solution, SymPy, machine theorem.

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1. Introduction

This supplementary monograph records the machine-verified Hilbert derivation theorem — the derivation layer of the whole programme (machine `sympy_derivation.py`, verdict `derivation_results.json`, LEDGER §3). The repository's slogan “direct solution from the classical Einstein equations” is only as strong as its derivation: this theorem is the machine proof that the equations the solvers march are exactly the Euler–Lagrange equations of the Hilbert action with a minimally coupled massless scalar.

The theorem anchors the entire tower programme: the books, the clocks and the chains of theorems No. 1–19 are statements about a specific system, and this monograph certifies that the system is not chosen but derived — from the action, in exact arithmetic, with the Roberts exact solution as the regression anchor. The whole derivation costs 1.9 s and runs in every CI pass.

2. Notation and Setting

The metric ansatz is the double-null form $ds^2 = -\alpha(u,v)^2 du dv + r(u,v)^2 d\Omega^2$; the action is $S = (1/2\kappa)\int \sqrt{-g} R + S_m$ with the scalar part $S_m = -1/2\int \sqrt{-g} (\nabla\Phi)^2$. The machine computes the Ricci scalar on the ansatz, the Hilbert stress-energy $T_{\{\mu\nu\}} = \Phi_{,\mu}\Phi_{,\nu} - (1/2)g_{\{\mu\nu\}}(\nabla\Phi)^2$, the d'Alembertian, and the Euler–Lagrange equations; $\kappa = 8\pi G$ in physics units, and the numerical code uses the Burko normalization $\kappa = 2$.

Table 20.1. The derivation objects (`derivation_results.json`)

Object	Symbolic value
Metric	$ds^2 = -\alpha(u,v)^2 du dv + r(u,v)^2 d\Omega^2$
Action	$S = (1/2\kappa)\int \sqrt{-g} R + S_m$, $S_m = -1/2\int \sqrt{-g} (\nabla\Phi)^2$
$\text{grad } \Phi^2$	$-4 \cdot \Phi_{,u} \cdot \Phi_{,v} / \alpha^2$
T_{uv}	0
T_{uu}	$\Phi_{,u}^2$
T_{vv}	$\Phi_{,v}^2$
$\text{box}\Phi$	$4(-r \cdot \Phi_{,uv} - \Phi_{,u} \cdot r_{,v} - \Phi_{,v} \cdot r_{,u}) / (\alpha^2 \cdot r)$

Object	Symbolic value
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3. Statement of the Theorem

Theorem 20 (the machine-verified Hilbert derivation). For the double-null ansatz above:

- (i) the stress-energy of the minimally coupled massless scalar is exactly $T_{uv} = 0$, $T_{uu} = \Phi_u^2$, $T_{vv} = \Phi_v^2$ — the null-null sector is empty, and the amplitude channels live on T_{uu} , T_{vv} ;
- (ii) the Hilbert identity holds exactly: the two nontrivial components of $\square\Phi$ coincide with $(\kappa/2)T_v$ after the Einstein equations are imposed (hilbert_identity components 0 and 1 nonzero, 2 and 3 identically zero);
- (iii) the Euler–Lagrange system solves term-by-term into the solver's equations: $C1_{r_{uu}} = -\kappa r \Phi_u^2/2 + 2\omega_u r_u$; $C2_{r_{vv}} = -\kappa r \Phi_v^2/2 + 2\omega_v r_v$; $UV_{r_{uv}} = (-e^{2\omega}/4 - r_u r_v)/r$ (and UV closed form identical); $TH_{\omega_{uv}} = -\kappa \Phi_u \Phi_v/2 - r_{uv}/r$;
- (iv) the checks pass: `R_phi_phi_spherical = true` (the spherical Ricci component), `roberts_solution_ok = true` with max residual 4.06×10^{-41} (the exact Roberts solution satisfies the derived system), `burko_E1_ok = true` (the Burko E1 normalization check);
- (v) the derivation is deterministic and cheap: `runtime_seconds = 1.9`, single SymPy pass, no numerical fitting anywhere.

4. Sketch of the Machine Proof

The machine builds the metric, computes $\sqrt{-g}$, the Christoffels, the Ricci tensor and scalar, the stress-energy and the d'Alembertian — all symbolically. The Einstein equations $G_{\{\mu\nu\}} = \kappa T_{\{\mu\nu\}}$ are formed and simplified; the two dynamical components project onto the $C1/C2/UV/TH$ forms of the solver after the $\omega = \ln \alpha$ substitution and the Burko normalization. The Hilbert identity is verified by simplifying $\square\Phi_v - (\kappa/2)T_v$ to zero on the solution manifold. The Roberts solution (exact α , r , Φ) is substituted into the derived

system; the max residual 4.06×10^{-41} is the mpmath evaluation of the simplified expressions.

Table 20.2. The solved system (system_solved layer)

Equation	Derived form
C1_r_uu	$-\kappa \cdot r \cdot \Phi_u^2/2 + 2 \cdot \omega_u \cdot r_u$
C2_r_vv	$-\kappa \cdot r \cdot \Phi_v^2/2 + 2 \cdot \omega_v \cdot r_v$
UV_r_uv	$(-e^{2\omega}/4 - r_u \cdot r_v)/r$
TH_ω_uv	$-\kappa \cdot \Phi_u \cdot \Phi_v/2 - r_{uv}/r$
UV_closed_r_uv	$(-e^{2\omega}/4 - r_u \cdot r_v)/r$ (identical to UV)

5. Numerical Results

Table 20.3. The checks layer (checks)

Check	Value	Status
R_phiphi_spherical	true	exact
roberts_solution_ok	true	residual 4.06×10^{-41}
burko_E1_ok	true	exact
runtime_seconds	1.9	—

The Roberts residual deserves its own sentence: 4.06×10^{-41} is the machine zero of the simplified symbolic evaluation — the exact solution satisfies the derived equations to the precision of the floating evaluation of exact expressions, not to a fitted tolerance. Together with the three-grid study of monograph No. 17 this separates the two error sources completely: the equations are exact, the discretization is second-order.

6. Corollaries

Corollary 20.1 (no hidden fields). The derivation uses exactly two fields (α or ω , r) plus the scalar Φ ; no auxiliary constructions enter. Every equation of every campaign machine is

therefore an equation about these three functions — the programme's ontology is fixed here.

Corollary 20.2 (the null-null emptiness is structural). $T_{uv} = 0$ identically explains why the uv-channel is carried by the geometric side (monograph No. 19): the amplitude book and the echo channel live on different components of one Hilbert tensor — a derivation-level fact, not a scheme choice.

Corollary 20.3 (the derivation is CI-pinned). At 1.9 s the whole derivation runs in every test pass together with the Roberts check; any tampering with the field equations breaks the identity and the CI simultaneously.

7. Honest Caveats

The derivation covers the Einstein–scalar system in double-null coordinates with a minimally coupled massless scalar; extensions (axion potentials, nonminimal coupling, a cosmological constant) are outside the theorem and would require a re-derivation. The Burko normalization $\kappa = 2$ is the code convention; the physics units are $\kappa = 8\pi G$, and the translation is linear. The Roberts check validates the derived system on one exact solution; it does not certify the solver's critical regime (that is monograph No. 17's business with its own caveats). Finally, the identity is verified after the Einstein equations are imposed — it is an on-shell identity, which is precisely the form the programme needs.

8. Reproducibility

Table 20.4. Reproduction protocol for Theorem 20

Step	Command	Result
1	<code>python3 sympy_derivation.py</code>	results/derivation_results.json (identity + system + checks)
2	<code>python3 roberts_test.py</code>	results/roberts_test.json (the discrete counterpart)
3	<code>python3 -m pytest tests/ -q</code>	58 passed (derivation tests included)

The machine runs in 1.9 s (deterministic, offline). The data correspond to commit 5b256cd; an autonomous copy lives in `einstein_direct/open9/machines/`.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/sympy_derivation.py`; `einstein_direct/results/derivation_results.json` (LEDGER §3).
- [2] Ibid.: `einstein_direct/roberts_test.py`; `einstein_direct/results/roberts_test.json` (the discrete validation).
- [3] Monographs No. 17, No. 19 of the theorems12 cycle.
- [4] D. Garfinkle. Choptuik scaling and the scale-invariant vacuum // (context of the double-null Einstein–scalar formulation).